

Bachelor's Thesis

**KAIZEN STRATEGIES FOR IMPROVING PLANNING AND PRODUCTION
FLOW IN PHARMACEUTICAL ENVIRONMENTS**

Double degree in Chemical Engineering and Industrial Electronics and Automation.

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Report and Annex

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Glossary

Eighty-twenty Rule Refers to the Pareto principle, which states that roughly 80% of effects come from 20% of the causes. [13](#)

Gemba A Japanese word meaning "the real place." In a KAIZEN™ context, it usually refers to the place where value is added, such as the shop floor. It can also refer to any place where work is being performed. [7](#)

JIT Just In Time. A system that "pulls" parts through production based on customer demand rather than "pushing" them based on projected demand. It results in line balancing and little to no excess inventory. [6, 7](#)

KAIZEN™ A Japanese term meaning "change for the better." It is a gradual, long-term approach that focuses on small, incremental changes to improve efficiency and quality. It was popularized by Masaaki Imai. [5–8, 10, 11](#)

KPI An English acronym for Key Performance Indicators, which are the key indicators of a given process or activity. [8, 9, 11](#)

Lean An English term meaning "no fat." In a management context, it is a strategy to reduce or eliminate waste or excess. [6, 7](#)

Muda A Japanese word for "Waste" and a key concept in TPS. It is one of three types of deviation from optimal resource allocation (Muda, Mura, and Muri). [6, 7, 11](#)

SMED Stands for **Single-Minute Exchange of Die**. A Lean manufacturing methodology for drastically reducing the time it takes to complete equipment changeovers. The goal is to make changeovers so quick that it becomes economically viable to produce smaller batch sizes, reducing inventory and improving production flexibility.. [13](#)

TPS Toyota Production System. A methodology that resulted from over 50 years of KAIZEN™ at Toyota. It is built on a foundation of Leveling with supporting pillars of Just-in-Time and Jidoka. [6](#)

Visual Management A set of techniques for creating a workplace that uses visual communication and control. The philosophy is "what gets measured and displayed, gets done." Simple tools are used to identify the target state and correct any deviations. [7](#)

VSA Stands for Value Stream Analysis. A Lean management method used to visually map and analyze the flow of materials and information required to deliver a product or service to a customer. VSA identifies value-added and non-value-added activities, highlights waste, and provides a basis for Kaizen (continuous improvement) efforts. It typically includes a current-state map, future-state vision, and an actionable plan to close gaps.. [12](#)

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5 Preface

5.1 The KAIZEN™ Philosophy

After World War II, Toyoda Kiichiro, then president of the Toyota Motor Company, saw the need to catch up with America to assure the survival of the automobile industry of Japan. Taiichi Ohno, Toyota's engineer, had a challenge: how to build a diversified line of products with the limited equipment Toyota possessed at the time. Instead of using batch production, he designed a system of integrated production that used one-piece flow, demonstrating that a trade-off between quality and productivity does not need to exist (Netland and Powell, 2016). Aligned with other practices, Toyota Production System (TPS) was born to defy the premise that bigger batches mean lower costs.

TPS, which ultimate goal is the absolute elimination of waste, is based in two pillars: just-in-time (JIT) and autonomation (automation with a human touch). The lack of the necessary capital resources to support high inventory levels was solved with the implementation of a just-in-time pull system. JIT means "to produce the necessary units at the necessary quantities at the necessary time". A company establishing this flow throughout can approach zero inventory. With autonomation, Toyota intended to create a built-in automatic checking system against small abnormalities. With this prevention mechanism, defective products were not produced. That was not the only advantage. While the machine is working properly, employees are liberated. Consequently, one worker could attend multiple machines, increasing production efficiency. Furthermore, autonomation came as a facilitator of improvement. Stopping the machine creates awareness and, through the total comprehension of the problem, improvement measures can be taken. (Ohno, 1988) Lean, based on TPS, is a methodology that accentuates customer needs, to improve quality and reduce costs through continuous improvement and employee involvement (Grabau, 2008). KAIZEN™, a Japanese word, refers to the concept of continuous improvement. It does not intend to be mistaken as innovation or disruption and elevated costs. On the other hand, KAIZEN™ is about small and subtle improvements. It is a low-cost low-risk process that assures incremental progress and sustainable changes in the long term.

At the beginning of the 21st century, Toyota Motor Company surpassed General Motors and became the world leader in automobile production, expanding and elevating the KAIZEN™ concept for its key role in the company's success. KAIZEN™'s methodology is based on five fundamental principles described in Massaki and Sanchez's book [1] :

- Create customer value

Value is what the customer is willing to pay for. Using a market-in approach to make informed decisions based on what the customers want and offer them that in the best way possible, improving customer experience.

- Create flow efficiency

Flow efficiency can be obtained through the elimination of three Ms: Muda (waste), Mura (unevenness) and Muri (overburden). The concept of Muda primarily originated from Taiichi Ohno's production philosophy in the early 1950s. He defined, in the industrial context he was inserted, seven sources of Muda: overproduction, waiting, inventory, motion, transportation, over-processing and defects. All the activities that do not add value to the process are considered Muda. Fujio Cho, former President of Toyota, defined waste as "anything other than the

minimum amount of equipment, materials, parts, space and worker's time, which are absolutely essential to add value to the product". Eliminating waste is the most effective way to increase productivity and decrease costs. Although **Muda** has gained bigger awareness, comprehending all three concepts allows for a greater understanding of **Lean** (Netland and Powell, 2016). **Mura** is the variation in a process that is not caused by the final customer, for example highs and lows in the planned production due to the production system or the irregular work rhythm. Leveling production using **JIT** can eliminate this irregularities, avoiding work spikes and long waiting times for workers. **Muri** means to overburden equipment or workers, demanding a faster rhythm for a longer period than expected. While in equipment it can provoke defects and failures, in people it can result in safety issues.

- Be **Gemba** oriented

Gemba is the place where value is added. Problems should be identified and solved there, at their root cause. Central to **KAIZEN™** is *genchi genbutsu* (go and see for yourself). Going to the **Gemba** means going to the place where the action really happens, to be able to thoroughly observe the reality of the processes.

- Empower people

Respect for people is a fundamental principle of the **KAIZEN™** philosophy. Kaizen recognizes that people are not the problem, processes are. It intends to improve every day, everywhere, with everyone, training all employees and encouraging people to think and share their own improvement ideas, systematically. It is in management's biggest interest to create the best possible conditions for everyone to do the best possible work.

- Be scientific and transparent

As important as identifying problems, is knowing how to explain and substantiate one's findings, supporting them with data. **Visual Management** is a basal tool of **KAIZEN™**, that highlights problems and allows for a quicker reaction to them.

6 The Problem at hand

6.1 Introducing the company

Now that we have introduced the base methodology of **KAIZEN™** we must understand the focus of this work. The project described has been implemented in a Pharmacology Company located on the Catalonia area. It focuses primarily on providing Contract Manufacturing and Packaging Services for Medicines and Dietary Supplements, both for Human and Veterinary use.

The facilities are equipped with state-of-the-art machinery and have a total area of 7.200 m², distributed across two facilities, one dedicated to the manufacturing of pharmaceuticals and the other to nutraceuticals.

In total the company has over 400 employees and about 80% of the workforce is established on the shop floor.

There are several manufacturing areas and work lines:

- TMP (Raw materials handling):

This section is where the mixture of active components, either liquids or solids, and ingredients are composed.

- PI (Compression):

In this section compression of solids into tablets occurs.

- PAC (Conditioning):

In this section the formula is conditioned. There are 3 forms of presentation:

Bottles: Only for liquids.

Jars: Containers of tablets for solids.

Pouches: For single use uncompress solids.

6.2 Scope and pilots

The scope of this project can be narrowed down into 3 **KAIZEN™ Events**, each one will be implemented in a different work line as a pilot.

Each **KAIZEN™** Event will be:

- Improvement of Line Efficiency.

Pilot: "Jars" Line

KPI: Line OEE

- Improvement of line Productivity

Pilot: "Liquids" Line

KPI: Line Productivity

- Improvement of planning

Pilot: Sourcing and production planning of PAC area.

KPI: OTIF

- Creating a system for daily management and continuous improvement culture.

Pilot: "Jars" Line

[KPI](#): Economic impact of meetings sourced improvement ideas.



7 Improvement of Line Efficiency

7.1 Objective Function

As in any optimization problem, we must first define the objective function. In the **KAIZEN™** methodology, improvement efforts are directed toward what is most valuable for the client, often summarized by the principles of GQCDM:

- Growth
- Quality
- Cost
- Delivery
- Motivation

In this particular **KAIZEN™** event, the focus will be on minimizing production cost. Specifically, this is pursued by maximizing the sustainable production capacity of the line.

$$\text{Objective function} = \max \left(\frac{\text{Total output units}}{\text{Time window}} \right) \quad (1)$$

In other words, equation 1 seeks to maximize the throughput rate, for example the number of orders the factory can process in a year.

This can also be expressed as:

$$\text{Objective function} = \max (\text{Rated line speed} \times OEE) \quad (2)$$

Here, the rated line speed represents the theoretical design capacity, while *OEE* adjusts for availability, performance, and quality losses. The introduction of this efficiency factor will be discussed in the following section.

It is evident that production output is strongly embedded in the cost structure of a company. A significant share of manufacturing cost is related to the utilization of machines and operators. By producing more in less time, the cost per unit decreases, thereby improving overall competitiveness.

7.2 OEE and profits calculation method

Overall Equipment Effectiveness (OEE) is a standard metric used to measure the efficiency of a production line. It identifies how much of the planned production time is truly productive. OEE is calculated by multiplying three core components: **Availability**, **Performance**, and **Quality**.

7.2.1 Availability

Availability measures the proportion of scheduled production time during which equipment is operational and ready for use. It accounts for planned downtime, such as maintenance or shift breaks, as well as unplanned losses caused by equipment breakdowns or delays.[2]

$$A = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \quad (3)$$

With that definition we first have to acknowledge 2 main terms:

1. Planned Time: The total scheduled production time.
2. Operating time: The Planned Time minus all planned stoppages like changeovers, cleaning, and scheduled breaks.

7.2.2 Performance

Performance evaluates whether equipment is running at its optimal speed during production. It captures losses due to reduced speed or minor stoppages that may not halt production but reduce efficiency.[2]

$$\mu = \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \times \text{Nominal Speed (units/h)}} \quad (4)$$

7.2.3 Quality

Quality assesses the proportion of defect-free products produced relative to the total output. This metric highlights losses due to rework or defective items, which reduce the overall effectiveness of the production process.[2]

$$Q = \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} \quad (5)$$

7.2.4 OEE calculation

By combining these three components, we get the final OEE formula, which shows the overall efficiency of the production line.

$$OEE = A * \mu * Q \quad (6)$$

An OEE score of 100% represents perfect equipment effectiveness, with no downtime, optimal performance, and zero defects.[3]

Or in fair terms:

$$OEE = \frac{\text{Operating Time (h)}}{\text{Planned Time (h)}} \times \frac{\text{Actual Units Produced (units)}}{\text{Operating Time (h)} \times \text{Nominal Speed (units/h)}} \times \frac{\text{Total OK unit output (units)}}{\text{Actual Units Produced (units)}} = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \times \text{Nominal Speed (units/h)}} \quad (7)$$

$$OEE = A * \mu * Q = \frac{\text{Total OK unit output (units)}}{\text{Planned Time (h)} \times \text{Nominal Speed (units/h)}} \quad (8)$$

7.2.5 Profit Calculation

Now that the concept of *OEE* is clear we could go back to equation 2. In **KAIZEN™** philosophy we focus on the non investment improvements. As we can infer changing the Nominal Speed of a line or machine could mean investing a huge amount of money in technology, instead of focusing on eliminating **Muda**. That's why we will focus on OEE as a **KPI** as the main metric for this project.

We can quantify the augmented capacity derived from an improvement in OEE as the reduced planned operating time of the line. In this work we will not be focused on improving sales directly, so the Total Number of units produced will be fixed. We won't focus on investing in new equipment, so the Nominal Speed of the line will stay constant. Every point of improvement of OEE will be inversely related to a decreased in the need for more planned time which in turns decreases the operating cost of labor of the line. Relating back to the objective function, equation 1.

7.3 Identifying improvement opportunities

A comprehensive analysis of Overall Equipment Effectiveness (OEE) provides a structured approach to identifying production inefficiencies. Following a [VSA](#), our project focused on the key opportunities identified to improve OEE. The project's baseline OEE was established after one month of observation in February, with the following results:

- Availability=41.6%
- Performance=66.2%
- Quality=99.89%
- OEE=27.5%

The baseline data clearly indicates that low availability is the primary factor limiting overall equipment effectiveness. This is further explained in Figure 1, which shows that only 40% of the available time is utilized for production.

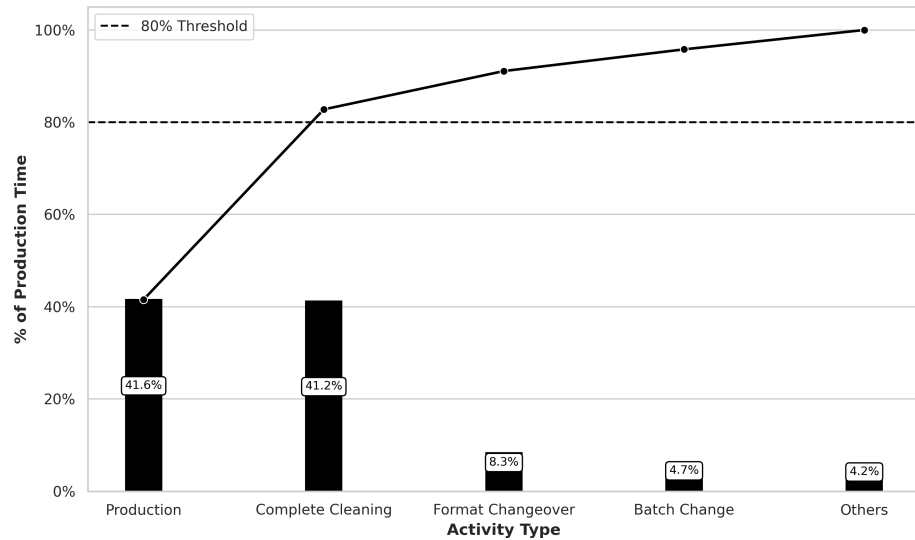


Figure 1: Pareto Chart of Available Production Time

The main reason for this loss in availability is the time dedicated to cleaning all machine equipment and the production space. This extensive cleaning is a direct consequence of stringent quality controls within the pharmaceutical industry, which mandate:

- The prevention of cross-contamination between different products or batches.

- The removal of product from machinery overnight to prevent quality degradation.

Figure 2, provides a more detailed breakdown of the availability losses. By applying [Eighty-twenty Rule](#) we can focus on the key areas for improvement.

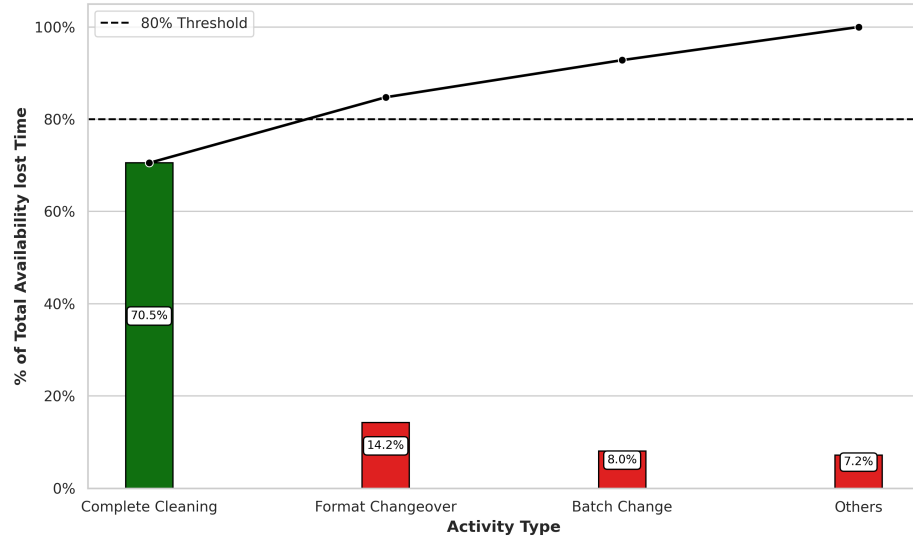


Figure 2: Pareto Chart of Main Losses on Availability. In Green Above 80% of cumulated time, in Red the rest 20% of time

The analysis confirms that a significant efficiency improvement can be achieved by reducing the time spent on cleaning. This objective will be addressed using the [SMED](#) Methodology.

7.4 Implementing the SMED Methodology

7.4.1 Brief introduction to the SMED

The [SMED](#) is a technique that helps us distinguish between Internal and External tasks as described in Dillon and Shingo [4] book *A Revolution in Manufacturing: The SMED System*. The core principle of the [SMED](#) system is to significantly reduce the time required for machine changeovers. This is achieved by converting as many setup operations as possible from *internal* tasks (which can only be performed when the machine is stopped) to *external* tasks (which can be completed while the machine is still running). The goal is to make all changeovers last less than ten minutes, thereby maximizing production time and enabling smaller, more flexible batch sizes.

This core principle of *internal* vs *external* tasks can be applied to multiple operations, not only limited to changeovers. In this case we will apply it to Cleanings. The basics steps described in Dillon and Shingo book are as follows:

1. Identify all tasks performed during the operation.
2. Identify all participants of the operation.
3. Distinguish between *internal* and *external* tasks.
4. Convert as much tasks from *internal* to *external*.

5. Distribute and balance internal tasks in order to minimize machine stopping time.
6. Standardize procedure, create checklists and [OPL](#).
7. Measure progress and identify bottle necks.

7.4.2 Brief introduction to the SMED

The [SMED](#) is a technique that helps us to significantly reduce the time required for machine changeovers and setups. Developed by Shigeo Shingo in the 1950s, the core principle is to convert as many setup operations as possible from *internal* tasks (those that can only be performed when the machine is stopped) to *external* tasks (those that can be completed while the machine is still running). The ultimate goal is to complete changeovers in less than ten minutes (a "single minute" in the literal sense of the word), thereby maximizing production time and enabling smaller, more flexible batch sizes.

As detailed in Shingo's foundational book, *A Revolution in Manufacturing: The SMED System* [4], this methodology is not limited to product changeovers but can be applied to any process with internal and external tasks, such as the cleaning procedure we have identified. The SMED implementation process follows these distinct steps:

1. **Preliminary Stage:** Identify and analyze the current setup or cleaning process. This involves documenting all tasks and their timings.
2. **Stage 1: Separate Internal and External Tasks:** Critically evaluate each step of the process and categorize it as either internal or external.
3. **Stage 2: Convert Internal to External Tasks:** This is the most crucial phase. Brainstorm and implement changes that allow internal tasks to be performed while the machine is running. For example, blowing off compressed air while the line is up.
4. **Stage 3: Streamline All Remaining Tasks:** Once the conversion is complete, focus on improving the efficiency of the remaining internal and external tasks. This involves using tools like checklists, visual aids, and improved tool organization. For instance, using quick-release clamps instead of bolts.
5. **Sustain and Standardize:** Create and maintain standardized work instructions, checklists and [OPLs](#) to lock in the improvements. Establish a system for measuring and tracking progress to identify future opportunities.

This structured approach ensures a systematic and measurable reduction in changeover or cleaning time, which directly increases equipment availability.

7.4.3 Documentation of tasks and timings

The first preeliminary step is to identify the task being performed by the operators and its timings.

A reduce list of this tasks is presented in the next table. But the original list can be found in the appendixes.

8 References

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